manifest reliably only over long periods. Such strategies are also unattractive for short-run investors, but investors with long horizons can afford to invest in them.

Viewed broadly, the opportunistic portfolio for long-run investing also represents the ability of long-run investors to profit from periods of elevated risk aversion or short-term mispricing.[20] In rational asset pricing models, prices are low because the average investor's risk aversion is high and investors bid down prices to receive high expected returns. If a long-horizon investor's risk aversion remains constant, then he can take advantage of periods with low prices. In behavioral frameworks, prices can be low because of temporary periods of mispricing. These can also be exploited by a long-term investor who knows that prices will return to fair values over the long run.[21] While the simple rebalancing strategy is counter-cyclical and has a value tilt, some of the best opportunistic strategies are even more counter-cyclical and strongly value oriented. Crises and crashes are periods of opportunity for truly long-run investors. As Howard Marks (2011), a well-known value investor, explains with great clarity: "The key during a crisis is to be (a) insulated from the forces that require selling and (b) positioned to be a buyer instead." That's what rebalancing forces the investor to do.

There have been debates in the academic literature on how large these hedging demand, long-run opportunistic effects really are. In a major paper, Campbell and Viceira (1999) estimate hedging demands to be very large that are easily double the average total demand for stocks by short-run investors. In Campbell and Viceira's model, the portfolio weight in equities for a long-term investor would have varied from –50% to close to 400% from 1940 to the mid-1990s. However, Brandt (1999) and Ang and Bekaert (2002), which appeared around the same time as Campbell and Viceira's paper, estimate small hedging demands. The long-run opportunistic demands depend crucially on how predictable returns are and the model used to capture that predictability. In chapter 8, I show that the evidence for predictability is weak, so I recommend that both the tactical and opportunistic portfolio weights be small in practice. Opportunistic hedging demands become much smaller once investors have to learn about return predictability or when they take into account estimation error.[22]

A system of predictable equity returns that has been widely studied in the portfolio choice literature is the Stambaugh (1999) system, where stock returns are driven by a valuation ratio like the dividend or earnings yield. The valuation ratio is a convenient instrument to capture time-varying expected returns. As dividends yields drop (or equity prices rise), future expected returns increase. The

[20] See Ang and Kjær (2011).

[21] See chapter 7 for a discussion of rational and behavioral determinants of risk premiums.

[22] See Brandt et al. (2005) and Pástor and Stambaugh (2012a).